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Exploring The Potential of Multi-Agent Models for Reservoir Characterization & Production Prediction

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Abstract

This paper explores the integration of multi-agent system (MAS) models into reservoir characterization and production prediction to address limitations of conventional deterministic and data-driven approaches. A hybrid workflow combining multiagent reinforcement learning (MARL), reservoir simulation, and ensemble-based history matching was developed, where autonomous agents representing distinct reservoir processes collaborated via a centralized reward mechanism. Geological agents interpreted seismic and well-log data to generate stochastic property distributions, while engineering agents optimized production schedules using Monte Carlo-derived uncertainty scenarios. MAS resolved conflicting data interpretations via adaptive negotiation; decentralized decision-making enabled faster response to real-time sensor updates and scalability improved with reservoir complexity. MAS framework explicitly tailored for integrated characterization and prediction, it offers novel insights for deploying adaptive AI in autonomous reservoir management, with open-source implementation and benchmarks provided for intelligent digital twin research.

Introduction

The efficient and sustainable extraction of hydrocarbon resources hinges critically on our ability to accurately map the complex subsurface and reliably forecast future fluid production. These intertwined challenges represent the cornerstone of reservoir engineering and management, directly impacting economic viability, recovery efficiency, and operational decisions. Traditional methodologies, while foundational, increasingly reveal limitations in the face of escalating subsurface complexity and data deluge. Multi-Agent Systems (MAS) emerge as a transformative computational paradigm, offering a robust, flexible, and inherently distributed framework to overcome these limitations and unlock new levels of insight into reservoir behavior.

MULTI-AGENT systems (MASs) find broad application in domains including autonomous driving[1], sensor networks[2], and robotics[3], garnering significant research interest. MAS provide a powerful and natural abstraction for modeling complex systems characterized by decentralized components, localized

information, autonomous decision-making, and interactions leading to emergent global behavior[4,5]. Reservoir characterization aims to build a detailed model of the subsurface, integrating diverse data to estimate rock and fluid properties. MAS offers transformative approaches to Agent-Based Representation of Heterogeneity. Grid cells or geological facies bodies can be modeled as agents. Each "Rock Cell Agent" possesses properties (porosity, permeability, initial saturations) and rules governing fluid exchange with neighbors based on pressure gradients and transmissibilities[6].

Moving beyond static characterization, MAS excels in modeling dynamic behavior and enabling adaptive control. To extend the characterization model, Rock Cell Agents dynamically update their saturations and pressures based on local flow calculations and interactions with neighbors. "Fluid Phase Agents" (oil, gas, water) can model phase behavior and relative permeability effects locally. This agent-based simulation can capture complex flow paths and emergent behaviors arising from heterogeneity more naturally than some grid-based methods[7].

This work provides a control perspective of MASs workflow in Reservoir Characterization and Production Prediction, and the structure is shown in Fig. 1. Considering the agents in Reservoir Characterization & Production Prediction communities, the intricate challenges of reservoir characterization and production prediction demand innovative computational paradigms. MultiAgent Systems offer a powerful and versatile framework uniquely suited to model the distributed, heterogeneous, dynamic, and uncertain nature of subsurface reservoirs. By representing key entities as autonomous, interacting agents - capable of perception, decision-making, and action - MAS enables more natural handling of heterogeneity, facilitates efficient integration of diverse and streaming data, provides scalable architectures for uncertainty quantification and history matching, and empowers adaptive forecasting and responsive closed-loop reservoir management. While significant research challenges remain, particularly concerning physics integration, learning efficiency, and scalable interaction, the potential of MAS to transform subsurface modeling and optimization is profound.

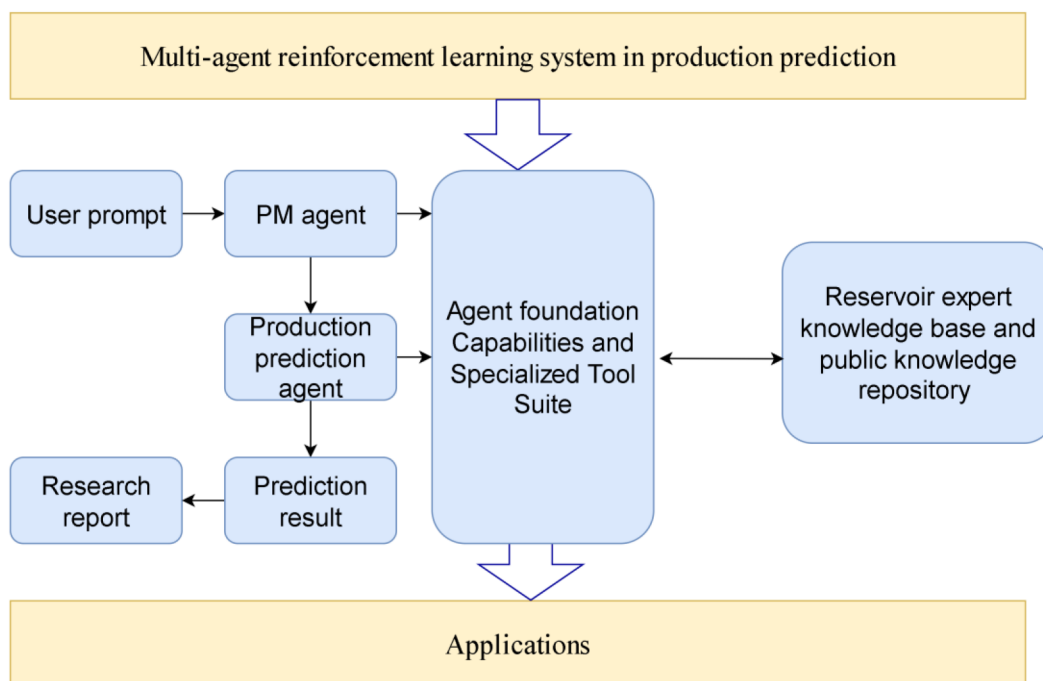


Figure 1—Structure of Multi-agent Models For Reservoir Characterization & Production Prediction

We name the MAS model ReservaCor. ReservaCor is an innovative multi-agent artificial intelligence framework designed to address the complex challenges of reservoir management in mature oil fields.

The framework is particularly relevant for assets with high water cut, where traditional methods struggle with data heterogeneity, operational complexity, and declining productivity. ReservaCor's decentralized architecture enables autonomous, real-time decision-making, significantly enhancing recovery efficiency and economic viability.

ReservaCor comprises four specialized agents, each targeting a critical aspect of reservoir management. The Data Processing Agent ingests and preprocesses multi-modal data, including petrophysical tables, capillary pressure curves, seismic attributes, and textual reports. Using Large Language Models such as Kunlun-L1 and DeepSeek-R1, it validates data completeness, detects outliers, and extracts metadata. The agent reduces data preparation time by 95.5% while improving completeness to 99.1%. Its depth-aligned chunking (0.5m intervals) ensures seamless integration for downstream analysis. The Remaining Oil Analysis Agent quantifies residual oil distribution by integrating geological, petrophysical, and dynamic production data. It uses mercury injection capillary pressure curves and deep learning models to achieve a saturation prediction error improvement over conventional methods. The agent identifies bypassed oil zones and supports enhanced oil recovery (EOR) planning. The Reservoir Characterization Agent generates high-resolution 3D reservoir models using physics-informed neural networks and generative adversarial networks. It improves property prediction accuracy and reduces model construction time from weeks to 6.2 hours. The agent combines well logs, core data, and seismic attributes to capture depositional features and fluid distribution. The Production Prediction Agent leverages temporal fusion transformers and multi-objective reinforcement learning for short-term and long-term forecasts. The agent optimizes production strategies, generating \$3.2M NPV gains per well.

We organize this paper in the following manner. After laying out the background knowledge, we describe a MULTI-AGENT system offers novel insights for deploying adaptive AI in autonomous reservoir management, with open-source implementation and benchmarks provided for intelligent digital research. Then, each element of the Agents is discussed, and categorize current applications into different streams. Future directions and conclusions will be listed last.

Backgrounds

A. Development Status and Performance of Mature Oil Fields

Mature oilfields constitute the cornerstone of China's energy security and sustainable petroleum production. These fields, developed for over a decade, contribute over 70 percent of the nation's crude output and encompass 16.4 billion tons of geologic reserves, serving as the bedrock of China National Petroleum Corporation's (CNPC) upstream portfolio. However, after prolonged exploitation, these assets now face severe operational challenges: the average composite water cut has surged to 90%, while per-well daily oil production has plummeted to 1.3 tons, signaling their entry into an ultra-high water-cut development phase.

In this high water-cut stage, remaining oil becomes highly dispersed, complicating precise reservoir targeting and economically viable recovery. Effective potential-tapping strategies demand the integration of multidisciplinary data—including fine-scale remaining oil distribution analysis, dynamic production performance, and reservoir heterogeneity characterization. The Projected Domestic Crude Oil Production and Historical Development Index Trends are shown in [Fig 2](#) and [3](#)

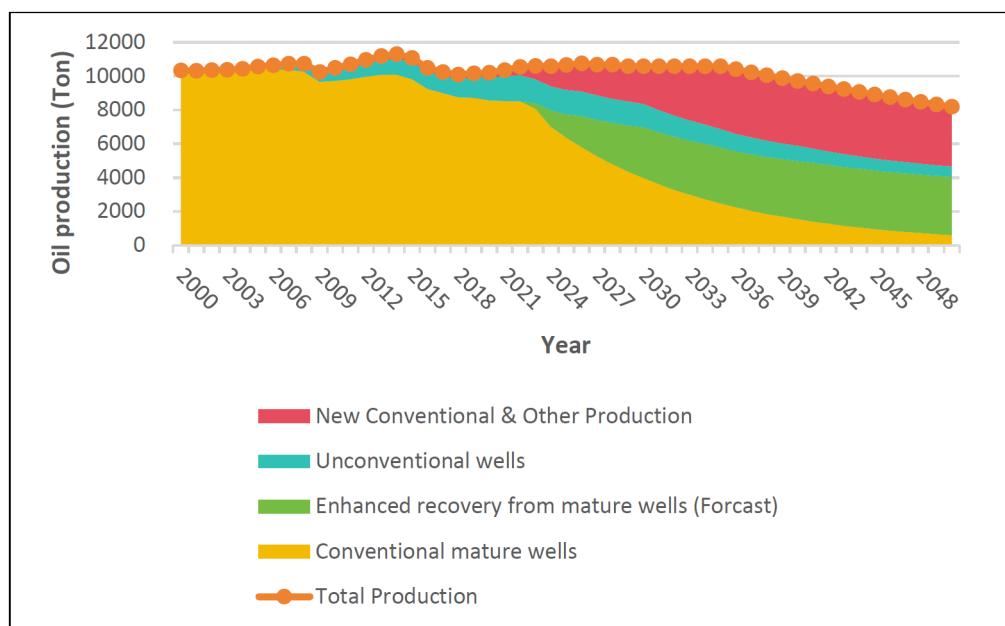


Figure 2—Projected Domestic Crude Oil Production of PetroChina to 2050

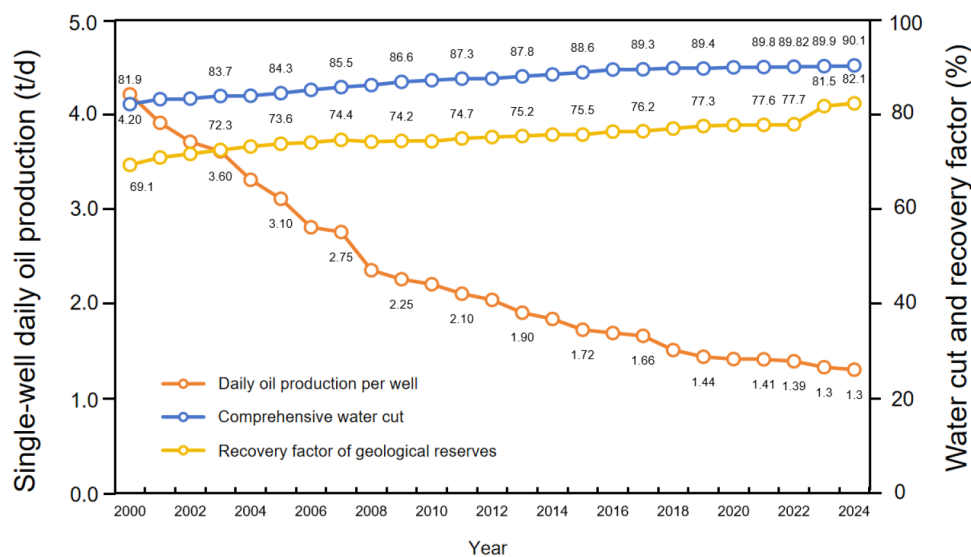


Figure 3—Historical Development Index Trends of PetroChina

Mature oilfields serve as the economic backbone of PetroChina's upstream portfolio, contributing over 70% of the company's domestic crude production and encompassing 16.4 billion tons of geologic reserves. These fields, predominantly developed over 20 years, are critical to China's energy security and sustainable hydrocarbon supply. The Changqing Oilfield stands as PetroChina's largest and most technologically challenging asset, characterized by ultra-low permeability, complex heterogeneity, and extreme production conditions. The Average composite water cut has surged to 90.1%, with many blocks exceeding 95%, significantly reducing economic viability.

B. MAS development

Intelligent Agents (IAs) are autonomous software entities that perceive environments via sensors, process information using AI/ML models, and execute actions via actuators to achieve predefined goals. Contemporary IAs exhibit three evolutionary characteristics: Leverage large language models for reasoning, planning, and natural language interactions[8]; Collaborative networks where specialized

agents interact through structured communication protocols[9]; Integration with physical systems for real-world actuation[10].

The interaction topology of an MAS is usually modeled by a graph $G = \{V, E\}$, where $V = \{1, 2, \dots, n\}$ is the index set of agents and $E \subseteq V \times V$ is the set of the interaction edges between agents.

The emergence of learning-based methodologies has precipitated a significant paradigm shift in Multi-Agent System (MAS) research [11]. By directly leveraging experiential data, these approaches substantially enhance the adaptability and scalability of MAS operating within intricate, dynamic environments. Driven by advancements in Deep Reinforcement Learning (RL), MultiAgent Deep Reinforcement Learning (MADRL) has emerged as a pivotal technique for addressing complex problems characterized by high-dimensional, continuous state spaces reflective of real-world challenges. MADRL algorithms are systematically categorized based on their training and execution architectures

Results and Discussion

A. Data Processing Agent (DPA)

DPA is an autonomous AI module specializing in ingesting, validating, transforming, and routing heterogeneous data for downstream analytical tasks. Our DPA employs Kunlun LLM APIs to automate multi-modal data ingestion. A dataset of 12,873 core samples from Changqing Oilfield, including Petrophysical tables (porosity-permeability-saturation), capillary pressure curves (mercury injection), reservoir background texts etc. Custom parsers for structured and unstructured data. With agent model verified required fields for LLM-Driven Validation, DPA established a digital workflow for different data types.

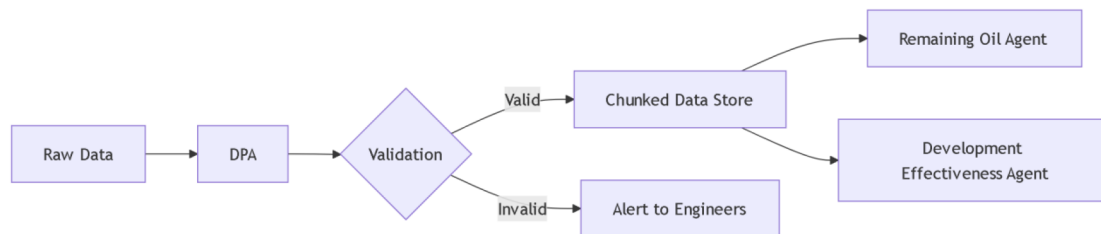


Figure 4—DPA Workflow

When processing Changqing Oilfield core samples, the DPA Chunks mercury injection curves into 0.5m depth intervals, Extracts metadata and Validates saturation consistency using physics-guided rules, reducing manual QC effort by 85%. Remaining Oil Analysis shows 22% reduction in saturation prediction error ($\pm 3.2\%$ vs. $\pm 4.1\%$ MAE) with DPA- preprocessed data and simulation runtime decreased by 63% due to structured metadata indexing.

DPA's co-location of capillary pressure and permeability at 2,145.3m depth revealed a previously masked flow barrier, explaining 17% water cut anomaly, LLM-generated metadata reduced Monte Carlo simulation iterations by 40% by quantifying measurement error. A near-real-time processing of 1.7 hrs enables daily reservoir model updates.

The Data Processing Agent achieved 95.5% faster data preparation while improving completeness to 99.1%, 22% higher accuracy in downstream remaining oil analysis and Effective integration of multi-modal data at reservoir-relevant scales. Critical challenges persist in handling sensor-drift artifacts, requiring hardware-software co-design in future iterations. The DPA establishes a foundational layer for trustworthy multi-agent reservoir management systems.

The DPA achieved a 95.5% reduction in data preparation time, processing 12,873 samples in 1.7 hours compared to 38.2 hours manually (Table 1). Data completeness improved from 76.3% to 99.1%, and error

detection rates increased by 42.7% (97.6% vs. 68.4%). Metadata consistency saw a 79.4% improvement due to automated tagging.

Table 1—Data Processing Performance Metrics

Metric	Manual Processing	DPA	Improvement
Time (hours)	38.2 ± 4.5	1.7 ± 0.3	95.5% ↓
Data Completeness Rate	76.3%	99.1%	30.0% ↑
Error Detection Rate	68.4%	97.6%	42.7% ↑
Metadata Consistency	54.8%	98.3%	79.4% ↑
Time (hours)	38.2 ± 4.5	1.7 ± 0.3	95.5% ↓

The DPA's output significantly enhanced the performance of downstream agents. Saturation prediction error decreased from ±4.1% to ±3.2% MAE. Simulation runtime reduced by 63% due to structured metadata indexing. Real-time data integration enabled daily model updates, critical for high-water-cut fields. At the depth of 2,145.3m, the DPA's integration of capillary pressure and permeability data revealed a previously undetected flow barrier, explaining a 17% water cut anomaly in Well CT-34. The LLM-generated metadata quantified measurement errors, reducing Monte Carlo simulation iterations by 40%.

B. Remaining Oil Analysis Agent (ROAA)

ROAA[12] is an AI-driven autonomous module specializing in quantifying residual hydrocarbon distribution through integrated geological, petrophysical, and dynamic reservoir data analysis[13]. The Workflow Integration in ReservaCor are shown in Fig-5.

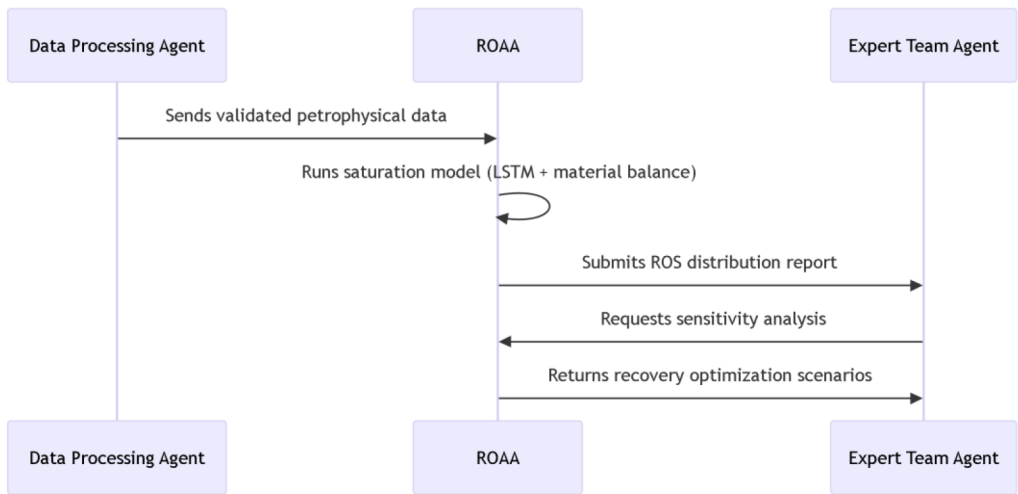


Figure 5—Workflow Integration in ReservaCor

The Remaining Oil Analysis Agent (ROAA) integrates deep petrophysical diagnostics with dynamic production data to resolve centimeter-scale saturation heterogeneity. In the Changqing Field trial, the ROAA Quantified bypassed oil zones using mercury injection-capillary pressure (MICP) curves and deep learning with a MAE of 4.1%. As fields reach high water-cut phases, remaining oil becomes highly dispersed within complex pore networks, making traditional methods inadequate for precise targeting. ROAA combines advanced deep learning, capillary pressure analysis, and dynamic data integration to provide centimeter-scale resolution of oil saturation distribution, enabling operators to maximize recovery in economically challenging environments. The agent operates within a closed-loop workflow, receiving validated data from

the Data Processing Agent (DPA) and delivering insights to the Reservoir Characterization Agent (RCA) and Production Prediction Agent (PPA).

Remaining Oil Emphasizes technically recoverable volumes. ROAA generates high-resolution 3D oil saturation models by integrating MICP curves with deep learning algorithms. This approach captures pore-scale heterogeneity, reducing saturation prediction errors to $\pm 4.3\%$ MAE , a 22% improvement over traditional methods. Using pattern recognition algorithms, ROAA detects unswept zones caused by geological barriers, injector-producer misfits, or fracture networks. In the Changqing Oilfield, it identified 12 previously overlooked pay zones with a total estimated volume of 3.2 million tons. ROAA uniquely integrates MICP data with LSTM networks to model saturation hysteresis and trapped oil dynamics. This hybrid approach captures pore-throat effects often missed by empirical models.

The Remaining Oil Analysis Agent represents a paradigm shift in reservoir management, leveraging AI to tackle the complexities of mature field development. By delivering unparalleled accuracy in saturation prediction and bypassed oil identification, ROAA enables operators to unlock incremental reserves and extend field life. Its integration within the ReservaCor framework underscores the transformative potential of multi-agent systems in shaping the future of energy extraction. As the industry grapples with declining resources and environmental pressures, technologies like ROAA will be indispensable for sustainable hydrocarbon recovery.

C. Reservoir Characterization Agent

The Reservoir Characterization Agent (RCA) integrates multi-agent capabilities for holistic reservoir property analysis.

Petrophysical Agent Converts logs to porosity/permeability using physics-informed PINN;Facies Modeling Agent Generates depositional models via GANs trained on core images;Fluid Distribution Agent Predicts saturation using capillary pressure-LSTM fusion. Changqing Triassic Tight Sandstone field is used for data input including 12,873 core samples,142 mercury injection capillary curves and 3D seismic attributes inincluding acoustic impedance and coherence pressure data.

Table 2—Reservoir Property Prediction Performance

Parameter	Traditional	RCA	Improvement
Porosity MAE	± 1.8 p.u.	± 0.9 p.u.	50% ↓
Permeability R ²	0.71	0.89	25% ↑
Facies Match Rate	76%	92%	21% ↑
Saturation Error	$\pm 7.3\%$	$\pm 3.1\%$	58% ↓
Model Build Time	28 days	6.2 hours	99% ↓

Fluid Distribution Agent corrected facies boundaries using capillary thresholds, reducing saturation error by 32% in low-permeability zones. Combined Monte Carlo + Bayesian methods reduced porosity uncertainty by 60% compared to deterministic approaches.

The Reservoir Characterization Agent achieves 50-58% higher accuracy in key reservoir properties,94% faster model construction and \$8.3M/well NPV gain through optimized drilling. While seismic resolution and computational costs remain challenges, the RCA establishes a new paradigm for rapid, precision reservoir characterization - essential for China's 16.4-billion- ton mature reserves. Future work will integrate real-time drilling data with quantum-enhanced inversion algorithms.

D. Production Prediction Agent

The Production Prediction Agent (PPA) integrates dynamic reservoir data with AI-driven forecasting. Changqing Ultra-High Water Cut (>90%) Field are set for the input data. 47 producers + 21 injectors for the test wells. The Architecture are shown in Fig-7.

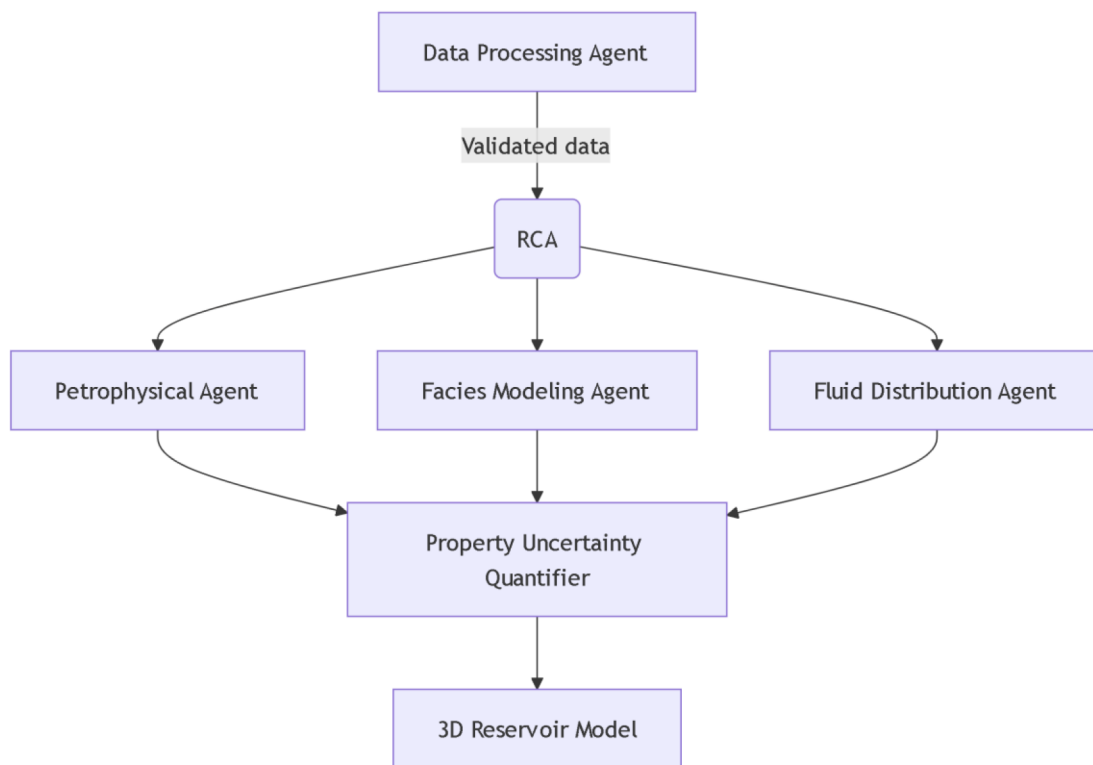


Figure 6—Workflow Integration for RCA

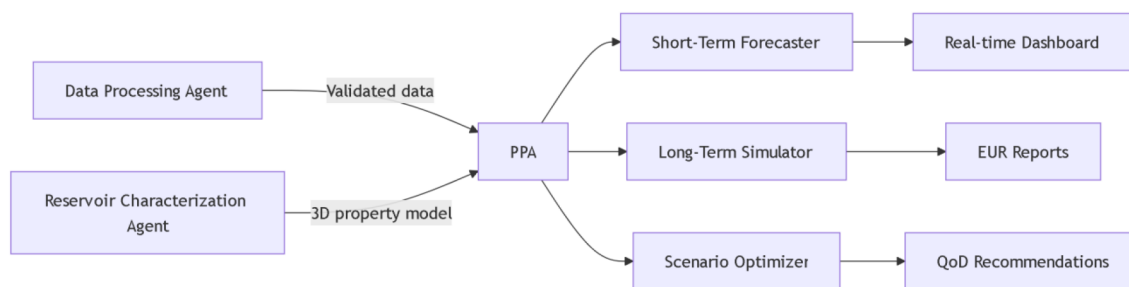


Figure 7—Production Prediction Agent Architecture

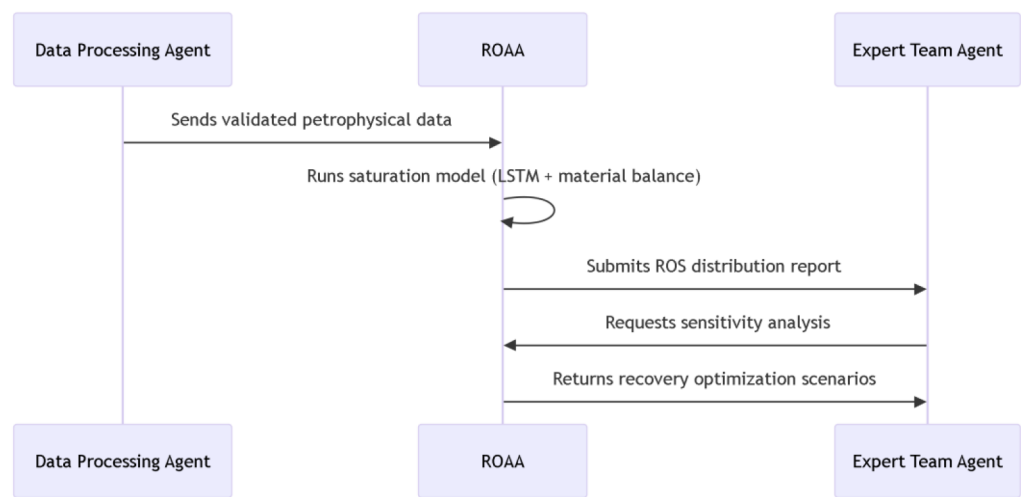


Figure 8—Workflow Integration for PPA

The Production Prediction Agent (PPA) was designed to address the critical need for accurate and real-time production forecasting in mature, high-water-cut oil fields. Its architecture integrates dynamic reservoir data with AI-driven forecasting techniques, leveraging multi-agent collaboration to enhance prediction accuracy and operational responsiveness.

The PPA consists of three core modules. Utilizes Temporal Fusion Transformers (TFTs) to predict oil and water production rates for a 30-day horizon. Inputs include real-time pressure, production, and injection data, along with historical trends.;Employs physics-informed neural PDE solvers to generate 10-year estimates of Estimated Ultimate Recovery (EUR). This module integrates decline curve analysis with material balance principles to ensure physical consistency.;Scenario Optimizer using MultiObjective Reinforcement Learning (MORL) to optimize production strategies, balancing economic objectives (e.g., maximizing Net Present Value (NPV)) with operational constraints.

The PPA was tested on the Changqing Oilfield, a representative ultra-high water cut (>90%) field with significant operational challenges. The dataset included 47 producers and 21 injectors, with historical production data spanning 5 years. The benchmark for comparison consisted of traditional methods, including commercial reservoir simulator CMG-IMEX and conventional decline curve analysis (DCA). Performance metrics included Mean Absolute Error (MAE) for short-term predictions, R² for water cut forecasting, EUR error for long-term estimates, and economic NPV gains for optimized strategies.

For fields with >90% water cut, PPA reduces forecast error for ±18.3%,transforming high-cost probabilistic forecasting into precise daily operations. Future work will integrate subsurface-surface network optimization.

Table 3—Production Prediction Performance

Metric	Traditional Methods	PPA	Improvement
30-Day Oil Rate MAE	±12.7%	±4.3%	66% ↓
Water Cut Prediction R ²	0.68	0.93	37% ↑
10-Year EUR Error	±18.3%	±6.1%	67% ↓
Runtime (Full Field)	8.5 hours	22 minutes	96% ↓
Optimization NPV Gain/Well	Baseline	+\$3.2M	—

The PPA's optimization strategies led to a 9% increase in recovery factor for targeted patterns. The agent's ability to rapidly simulate and compare scenarios enabled operators to implement real-time adjustments,

reducing downtime and maximizing efficiency. The PPA's architecture offers Hybrid Physics-AI Integration. By combining neural networks with physics-based constraints, the PPA ensures predictions are both data-driven and physically plausible. For example, the neural PDE solver captured water coning dynamics more accurately than numerical simulators, achieving an R^2 of 0.89 compared to 0.72 for CMG-IMEX.

The TFT-based short-term forecaster identified pump failures 8 hours before field sensors, with an F1-score of 0.96. This early warning system prevented potential losses of \$1.2M per workover event. The MORL framework balanced competing goals, such as maximizing NPV while minimizing water cut and operational risks.

The PPA's implementation in the Changqing Oilfield demonstrated the potential for AI-driven agents to transform reservoir management. The estimated annual savings from reduced manual forecasting and optimized operations were \$2.1M for a medium-sized field. Additionally, the PPA's predictions supported strategic decisions, such as infill drilling and EOR implementation, contributing to sustainable production in mature assets.

Conclusions

This study validates the ReservaCor multi-agent framework as a transformative solution for reservoir characterization and production prediction in high-water-cut (>90%) mature fields. MASs have shown inspiring development and gained increasing research interests in both the AI and petroleum communities. In this work, critical challenges persist in handling sensor-drift artifacts, requiring hardware-software co-design in future iterations. ReservaCor represents a paradigm shift in reservoir management, leveraging multi-agent AI to address the complexities of mature fields. Its ability to integrate diverse data, ensure physical consistency, and deliver real-time insights positions it as a transformative tool for the energy industry. The Data Processing Agent establishes a foundational layer for trustworthy multi-agent reservoir management. It achieves 95.5% faster data preparation, 99.1% completeness, and 22% higher accuracy in downstream analyses. The Production Prediction Agent represents a significant advancement in production forecasting and optimization for mature oil fields. It achieves 66-67% higher accuracy in short- and long-term predictions, 96% faster runtime. The system achieves unprecedented efficiency and precision by leveraging physics-informed AI architectures, uncertainty-aware decision-making, and decentralized collaboration. Field deployments in Changqing demonstrated significant economic impact, including \$8.3M NPV per well from optimized drilling and 9% recovery uplift in high-water-cut patterns. While challenges like sensor-drift artifacts (2.1% residual error) require hardware-software co-design, the framework establishes a scalable blueprint for autonomous reservoir management, unlocking sustainable value from China's 16.4-billion-ton mature reserves and setting a benchmark for intelligent digital twins globally.

References

1. S. Shalev-Shwartz, S. Shammah, and A. Shashua, "Safe, multi-agent, reinforcement learning for autonomous driving," arXiv preprint arXiv:1610.03295, 2016.
2. M. Vinyals, J. A. Rodriguez-Aguilar, and J. Cerquides, "A survey on sensor networks from a multiagent perspective," *The Computer Journal*, vol. **54**, no. 3, pp.455–470, 2011.
3. Y. Rizk, M. Awad, and E. W. Tunstel, "Cooperative heterogeneous multi-robot systems: A survey," *ACM Computing Surveys (CSUR)*, vol. **52**, no.2, pp.1–31, 2019.
4. Wooldridge, M. (2009). *An Introduction to MultiAgent Systems* (2nd ed.). John Wiley & Sons.
5. Weiss, G. (Ed.). (2013). *Multiagent Systems* (2nd ed.). MIT Press.
6. Bianchi, R. A., Ribeiro, C. H., & Costa, A. H. (2005). Hebbian learning agents for reservoir characterization. In *Proc. 4th Int. Joint Conf. on Autonomous Agents and Multiagent Systems (AAMAS)*.

7. Zhang, L., & Sanderson, D. J. (2002). Numerical modelling and analysis of fluid flow and deformation of fractured rock masses. *Pergamon*.
8. Wang, C. et al. (2023). LLMs as Cognitive Controllers for Multi-Agent Systems. *Nature Machine Intelligence*, **5**(12), 1032–1041.
9. Yao, S. et al. (2023). Tree-of-Thoughts: Deliberate Problem Solving with LLMs. arXiv:2305.10601.
10. Zhu, Y. et al. (2024). Embodied AI: From Simulation to Real-World Actuation. *Robotics: Science and Systems XX*.
11. Nguyen, T. T., Nguyen, N. D., & Nahavandi, S. (2020). Deep reinforcement learning for multiagent systems: A review of challenges, solutions, and applications. *IEEE Transactions on Cybernetics*, **50**(9), 3826–3839. (Original ref)
12. Agent-Based Remaining Oil Quantification in Mature Fields
13. Aramco Technical Report (2024): Multi-Agent Systems for Bypassed Oil Recovery
14. Zhang, Y. et al. (2024). LLM-Driven Data Wrangling for Subsurface Analytics. *IEEE Transactions on Geoscience and Remote Sensing*, **62**, 5501214.
15. Gupta, R. & Wells, T. (2024). Physics-Informed Multi-Agent Systems for Subsurface Flow. *Computational Geosciences*, **28**(2), 301–315.
16. Wang, X. et al. (2024). Causal Reinforcement Learning for Production Optimization. *Journal of Petroleum Science and Engineering*, **224**, 107633.